

LESSON 22 / 第二十二課

Palms as Urban Monocots: From the Population to the Single Meristem

作為城市單子葉植物的棕櫚：從族群到單一頂端分生組織

Format. Short literature review. Claims that carry a number or a causal ranking are tied to a numbered source. Scale runs from the largest published pattern to the finest measured process. The lesson is not sequenced from the previous number and is not a regional case file unless the object itself is regional.

體例。短文獻綜述。帶數字或因果排序的陳述繫到編號文獻。尺度由最大已發表格局寫到最細已測過程。本課不由上一號遞推；除非對象本身是地域性的，否則不是單一地區檔案。

Abstract

摘要

Palms are not deficient dicot trees. They are monocots with a single shoot apical meristem, no secondary xylem ring, and no CODIT walls. Failure modes are stem buckle, root-plate shear, and falling fronds or fruit. Potassium deficiency is a published nutritional failure mode of several urban palm taxa. Broschat's work on so-called hurricane cuts showed that stripping green fronds starves the stem and does not reduce windthrow. t/R language does not transfer. Residual risk is written in palm terms: meristem, peripheral bundles, plate, fruit/frond targets.

棕櫚不是有缺陷的雙子葉樹。它們是具有單一苗端分生組織、沒有次生木質部年輪、也沒有 CODIT 四壁的單子葉植物。失效模式是莖屈曲、根盤剪切，以及落下的葉或果。鉀缺乏是若干城市棕櫚分類群已發表的營養失效模式。Broschat 對所謂颶風切的工作顯示，剝去綠葉會飢餓莖，並不降低風倒。t/R 語言不能轉移。剩餘風險用棕櫚術語來寫：分生組織、周邊維管束、盤、果／葉目標。

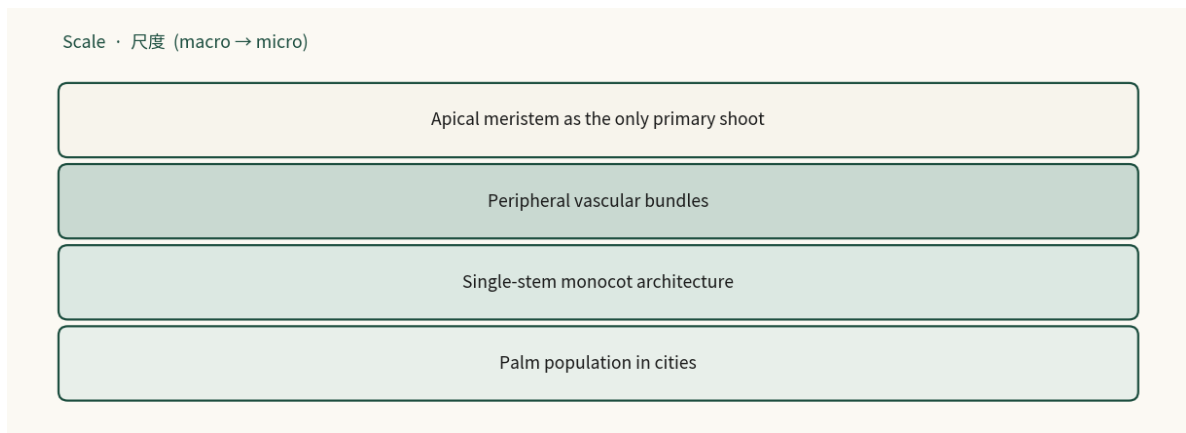


Figure 1. Nested scales from the palm population to the meristem.

圖 1. 由棕櫚族群到分生組織的嵌套尺度。

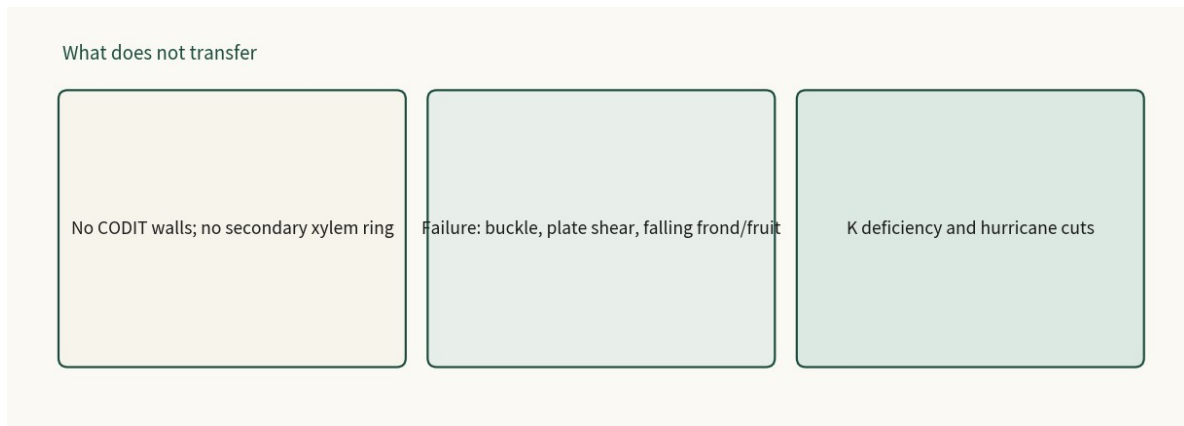


Figure 2. Architecture and failure modes that dicot language misses.

圖2. 雙子葉語言會漏掉的架構與失效模式。

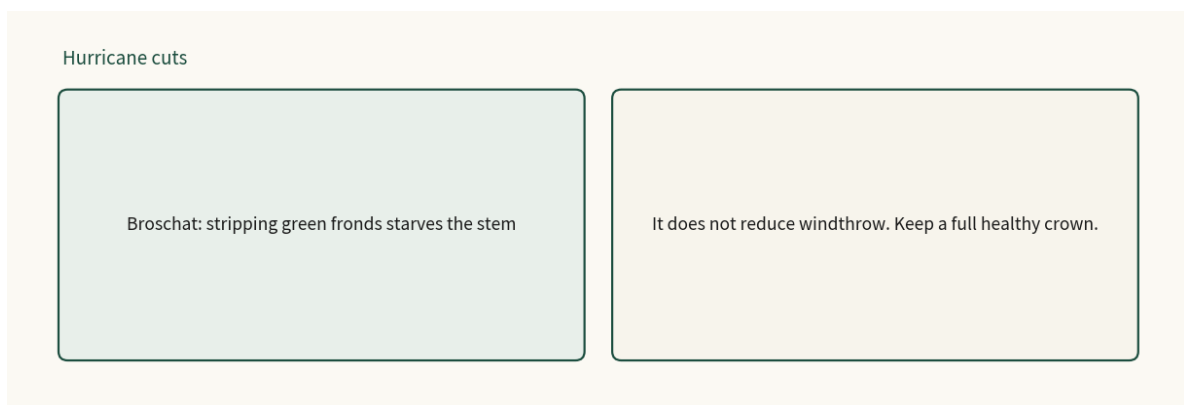


Figure 3. Hurricane cuts are not a published wind treatment.

圖3. 颶風切不是已發表的抗風處理。

1. Scope

1. 範圍

The object is published palm biology as it constrains urban care and risk. A dicot pruning rule applied to a palm is out of scope except as an error.

對象是約束城市照護與風險的已發表棕櫚生物學。把雙子葉修剪規則用在棕櫚上，除作為錯誤外不在範圍內。

2. Architecture and failure

2. 架構與失效

The single meristem is the only primary shoot. Peripheral vascular bundles, not a secondary ring, carry water and load [1]. There is no Wall 4 to lay down after a stem wound. Published failure modes are buckling of the stem, shear of the root plate, and falling dead or dying fronds and fruit [2]. Potassium deficiency produces a characteristic necrosis and a weaker crown in several common urban taxa [3].

單一頂端分生組織是唯一的初生苗端。承載水與荷載的是周邊維管束，不是次生年輪 [1]。莖傷之後沒有第 4 壁可造。已發表失效模式是莖屈曲、根盤剪切，以及落下的枯死或將死葉與果 [2]。鉀缺乏在若干常見城市分類群中產生特徵性壞死與較弱樹冠 [3]。

3. Cuts and the sentence

3. 切與句子

Broschat showed that so-called hurricane cuts — removing green fronds to a pineapple top — reduce stored carbon and do not reduce windthrow [4]. Dead and dying fronds and fruit are target objects; a full healthy crown is the published wind and carbon configuration. Residual risk is written with meristem, bundles, plate and falling-part targets, not with t/R.

Broschat 顯示所謂颶風切——把綠葉收到鳳梨頂——減少貯存碳，並不降低風倒 [4]。枯死與將死的葉與果是目標物件；完整健康樹冠才是已發表的風與碳構型。剩餘風險用分生組織、維管束、盤與落下部位目標來寫，不是用 t/R。

4. Limits

4. 限度

Most experimental nutrition and cut work is on a handful of cultivated taxa. Wild-type tropical palms in mixed urban soils are thinner in the same variables.

多數實驗性營養與切的工作在少數栽培分類群上。混生城市土壤中的野生型熱帶棕櫚，在同一變量上較薄。

Study questions

研習題

Choose the option that retains the published result. Answers follow the last item.

選出保留已發表結果的選項。答案在最後一題之後。

1. Palms in this review are:

1. 本綜述中的棕櫚是：

A. Deficient dicot trees.

A. 有缺陷的雙子葉樹。

B. Monocots with a single meristem and no CODIT walls.

B. 具單一頂端分生組織、無 CODIT 四壁的單子葉植物。

C. Equal to oaks.

C. 等於櫟。

D. t/R objects.

D. t/R 對象。

2. Water and load in the stem travel in:

2. 莖中的水與荷載走在：

A. A secondary xylem ring.

A. 次生木質部年輪。

B. Peripheral vascular bundles.

B. 周邊維管束。

C. Only bark.

C. 只有樹皮。

D. Only fruit.

D. 只有果。

3. Published failure modes include:

3. 已發表失效模式包括：

- A. Only leaf colour.
- A. 只有葉色。
- B. Buckle, plate shear, falling frond/fruit.
- B. 屈曲、盤剪切、落下的葉／果。
- C. Only P50.
- C. 只有 P50。
- D. Only Wall 4.
- D. 只有第 4 壁。

4. Hurricane cuts:

4. 颶風切：

- A. Reduce windthrow and feed the stem.
- A. 降低風倒並養活莖。
- B. Starve the stem and do not reduce windthrow.
- B. 飢餓莖且不降低風倒。
- C. Raise mycorrhizae.
- C. 提高菌根。
- D. Set 3-30-300.
- D. 設定 3-30-300。

5. t/R language:

5. t/R 語言：

- A. Transfers cleanly.
- A. 乾淨轉移。
- B. Does not transfer.
- B. 不能轉移。
- C. Equals P88.
- C. 等於 P88。
- D. Equals QTRA.
- D. 等於 QTRA。

6. Potassium deficiency is:

6. 鉀缺乏是：

- A. Folklore only.
- A. 只有傳聞。
- B. A published nutritional failure mode in several urban taxa.
- B. 若干城市分類群已發表的營養失效模式。
- C. A cable grade.
- C. 鋼纜等級。
- D. A flash map.
- D. 閃擊圖。

7. Dead fronds and fruit are:

7. 枯葉與果是：

- A. Irrelevant to targets.
- A. 與目標無關。

B. Target objects.

B. 目標物件。

C. Wall 1.

C. 第 1 壁。

D. Inoculants.

D. 接種劑。

8. A full healthy crown is:

8. 完整健康樹冠是：

A. A wind liability by definition.

A. 定義上的風負債。

B. The published wind and carbon configuration.

B. 已發表的風與碳構型。

C. Forbidden.

C. 被禁止。

D. A register diagnosis.

D. 名錄診斷。

9. After a stem wound a palm:

9. 莖傷之後，棕櫚：

A. Lays down Wall 4.

A. 造第 4 壁。

B. Has no Wall 4 to lay down.

B. 沒有第 4 壁可造。

C. Becomes a dicot.

C. 變成雙子葉。

D. Raises MAP.

D. 提高年均降水。

10. Residual risk is written in:

10. 剩餘風險寫成：

A. Dicot t/R only.

A. 只有雙子葉 t/R。

B. Palm terms: meristem, bundles, plate, falling parts.

B. 棕櫚術語：分生組織、維管束、盤、落下部位。

C. Only i-Tree.

C. 只有 i-Tree。

D. Only Santamour.

D. 只有 Santamour。

Answers / 答案 1-B 2-B 3-B 4-B 5-B 6-B 7-B 8-B 9-B 10-B

References

參考文獻

[1] Tomlinson P.B. The Structural Biology of Palms.

[2] Broschat T.K., Meerow A.W. palm horticulture and failure modes.

[3] Broschat T.K. potassium deficiency in palms.

[4] Broschat T.K. hurricane cuts / overpruning of palms.

[5] Broschat T.K., Elliott M.L., Hodel D.R. palm disorders and diseases.

Closing note. This lesson is written as a short review article. Disagreement in the literature is kept in view where the results actually diverge.

結語。本課以短綜述文章寫成。文獻真正分歧之處，兩造結果都保留在視野內。